

# Reading Support: Abstraction and the Basics

Companion to Week 1

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# Abstraction begins with a purpose

**Supports:** Mathematics begins by deciding what to ignore

*When we abstract, we attend to some properties of things and leave others aside. The result is useful only in relation to a question. A string, a tree, and a function can all represent the same linguistic material while preserving different relations in it.*

*Mathematics studies what remains invariant across the cases admitted by an abstraction. It therefore requires us to say both what varies and what we intend to preserve.*

MOLR Preface; §1

# A set is determined only by its members

**Supports:** A set records membership, not order or repetition

*A set is a collection regarded as one object. Writing  $\{a, b, c\}$  says that  $a$ ,  $b$ , and  $c$  belong to that collection. Their written order contributes nothing to the set, and listing a member more than once does not create another member.*

*Thus  $\{0, 2, 4, 126, 8, 1040, 0\}$  and  $\{0, 2, 4, 8, 126, 1040\}$  name the same set. This convenience becomes important when the members cannot usefully be listed one by one.*

MOLR §1.1

# A condition may describe the members

**Supports:** Set-builder notation states the admissibility condition

*Set-builder notation has the form  $\{x \mid P(x)\}$ : the set of all objects  $x$  for which condition  $P$  holds. It does not assume that the set is actually written out or even conveniently listable.*

*The vertical stroke can be read as “such that.” The expression therefore states an admissibility condition for membership rather than a procedure for finding the members.*

MOLR §1.1

# Empty objects must be kept distinct

**Supports:** The empty set is not a set containing the empty set

*The empty set  $\emptyset$  has no members. The singleton  $\{\emptyset\}$  has one member, namely the empty set. Consequently,  $\{\emptyset\} \neq \emptyset$ .*

*The power set  $\mathcal{P}(A)$  contains every subset of  $A$ , including  $A$  itself and  $\emptyset$ . If  $A = \{a, b\}$ , then  $\mathcal{P}(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$ .*

MOLR §1.1

# Strings add order and multiplicity

**Supports:** Strings restore order and repetition

*A string is a finite sequence of elements from an alphabet. Unlike a set, it distinguishes  $ab$  from  $ba$ , and  $aa$  from  $a$ . The empty string  $\epsilon$  has length zero.*

*If  $A$  and  $B$  are sets of strings, their concatenation is  $AB = \{xy \mid x \in A, y \in B\}$ . The Kleene star  $A^*$  contains every finite concatenation of members of  $A$ , including  $\epsilon$ .*

MOLR §1.2

# Relations select ordered pairs

**Supports:** A relation admits selected pairs

*A relation  $R$  from  $A$  to  $B$  is a subset of  $A \times B$ . Writing  $aRb$  abbreviates  $(a, b) \in R$ . Relations may be reflexive, symmetric, antisymmetric, or transitive, but none of these properties follows merely from being a relation.*

*The child-of and sister-of relations on a tree preserve different facts. The choice of relation determines which structural questions the representation can answer.*

MOLR §1.3

# Functions restrict relations

**Supports:** A function is a relation with a stricter condition

*A function from  $A$  to  $B$  assigns every member of its domain exactly one value in its codomain. Several inputs may share an output, but one input cannot have two outputs under the same function.*

*The notation  $B^A$  denotes the set of all total functions from  $A$  to  $B$ . It treats functions themselves as values that can enter further relations and functions.*

MOLR §1.3

# A tree can be viewed through parenthood

**Supports:** The same tree supports relations and functions

*A tree has nodes and links expressing parenthood. Every nonroot node has exactly one parent, there is one overall root, and the structure is connected. Leaves terminate paths because they have no children.*

*Parent-of can therefore be represented as a function on the nonroot nodes. Child-of cannot generally be a function on parents, because a parent may have more than one child.*

MOLR §1.3, "Trees as functions"

# Composition depends on matching domains

**Supports:** Composition connects matching domains

*If  $g : A \rightarrow B$  and  $f : B \rightarrow C$ , their composition  $f \circ g : A \rightarrow C$  is defined by  $(f \circ g)(x) = f(g(x))$ . The output space of  $g$  supplies admissible inputs to  $f$ .*

*Composition is associative whenever the relevant domains connect. This does not make it commutative:  $f \circ g$  and  $g \circ f$  may differ or one of them may not even be defined.*

MOLR §1.3, “Function composition”

# A commuting diagram makes equality of routes visible

**Supports:** A diagram asks whether two routes agree

*A diagram of functions displays objects and the mappings that connect them. It commutes when every directed route with the same starting and ending objects yields the same result.*

*Commutativity is therefore a substantive equality, such as  $h = g \circ f$ . The diagram does not guarantee it. It gives us a compact way to ask whether independently described operations preserve the same relation.*

MOLR §1.3, “Diagrams of functions”